



PIYUSH GIRISH KULKARNI

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PROFILE SUMMARY

Versatile data and AI professional with a strong computer science foundation and years of experience across Data Engineering, AI/GenAI, and Data Science. Proven track record building production ETL/ELT pipelines, leading technical teams, deploying semantic search engines, and delivering data-driven solutions across cloud platforms.

SKILLS

- Data Engineering: Apache Airflow, Dagster, Apache Kafka, DuckDB, ETL/ELT, Medallion Architecture, AWS S3.
- AI & GenAI: LangChain, RAG, Self-RAG, PyTorch, Sentence Transformers, LLMs, MCP Servers.
- Data Science & BI: Pandas, NumPy, Scikit-learn, Power BI, Streamlit, EDA, Deep Learning.
- Core & Cloud: Python, SQL, JavaScript, Git, CI/CD, AWS, Azure, Docker, PostgreSQL, MySQL, REST APIs.

WORK EXPERIENCE

Data Engineer (Working Student) | STATISTA GmbH – Hamburg, Germany | *July 2025 – Current*

- Built automated ETL pipelines using Selenium and REST APIs to centralize 20+ international economic datasets into a scalable AWS S3 data lake with medallion architecture.
- Optimized system stability by implementing chunked processing in pandas to ingest files containing millions of records
- Engineered comprehensive data quality validations and unit mapping transformations to standardize global data formats.

GenAI & Agentic AI Engineer (Company Project) | Galagos AI – Hamburg, Germany | *March 2025 – August 2025*

- Led a 5-member team to architect a Semantic Search Engine using LangChain and ChromaDB for complex bioinformatics queries.
- Developed an autonomous Self-RAG agent to iteratively retrieve and evaluate context, significantly improving search accuracy.
- Directed core architectural decisions for automated data ingestion pipelines, vector database design, and external platform integrations via MCP servers.

Data Scientist | Quantum Innotek Solutions – India | *Dec 2023 – Aug 2024*

- Engineered a targeted search and matching algorithm using KNN and FLAN to process complex data structures and deliver relevant results.
- Built custom relational databases and data access layers via SQLAlchemy and MySQL with Flask backends.
- Developed and maintained Git-based CI/CD pipelines for automated continuous deployment to Linode production servers.

Software Engineer & Data Analyst | CIS IT Solutions Pvt Ltd – India | *Feb 2023 – Dec 2023*

- Engineered automated cloud workflow applications using AWS and Microsoft Power Apps to streamline client business pipelines.
- Conducted deep EDA and deployed interactive dashboards in Power BI for operational research
- Developed predictive data models and automated mathematical logic using Python and NumPy

EDUCATION

SRH University of Heidelberg Campus Hamburg, Hamburg, Germany

Master of Science – Applied Data Science & Analytics (Oct 2024 – Present)

- Relevant Coursework: Machine Learning, Deep Learning, Big Data with Python, Statistics, Data Analysis, Data Management.

MIT ADT University, Pune, India

Bachelor of Technology – Computer Science & Engineering (Class of 2022)

- Relevant Coursework: Core AI Technologies, Deep Learning, Machine Learning.
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ACADEMIC PROJECTS

FoodVisionAI-Food Recognition & Nutrition App - [Link](#)

- Vision Architecture: Fine-tuned PyTorch CNNs (EfficientNet) on a large-scale dataset of 131K+ images to achieve a 95% real-time dish classification accuracy.
- Semantic Matching Engine: Utilized Sentence Transformers to accurately map visual predictions to a 1,000+ item nutrition database with an ultra-low query latency of ~14ms.
- GenAI Deployment: Built a responsive Streamlit web application integrating local LLMs (Ollama) to deliver interactive AI-driven Q&A and comprehensive food descriptions.

Topic Modelling Using LDA (July 2021 to May 2022) - [Link](#)

- NLP Framework: Utilized Latent Dirichlet Allocation (LDA) frameworks to rapidly process, cluster, and analyze extensive corpora of unstructured biomedical text data.
- Summarization Architecture: Built a specialized question-answering model leveraging the PICO framework for precise context extraction from dense COVID-19 datasets.
- Text Processing Pipelines: Engineered automated regex, text tokenization, and vectorization pipelines to transform raw literature into structured machine-readable formats.

IoT Smoke Detection Data Pipeline- [LINK](#)

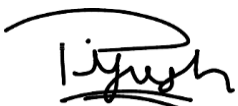
- ML Integration: Developed and deployed automated Scikit-learn predictive models via REST APIs to handle real-time classification inference.
- Streaming Infrastructure: Built a live event-streaming and batch pipeline using Apache Kafka, Docker, and Airflow to stream high-frequency IoT sensor metrics.
- Model Observability: Integrated Prometheus and Grafana for live data drift monitoring and achieved 95% automated test coverage across the model environment.

ACHIEVEMENTS

- AWS Certified Cloud Practitioner
- Copyright: [Perspective Approach for Mango Ripening Classifiers](#)

LANGUAGES

- English – Native
- German – Conversational (A2 – B1)



Place: Hamburg, Germany
Date: 26-06-2026